

GOVERNMENT OF INDIA — MARITIME ENVIRONMENTAL PROTECTION
DIVISION
HEADQUARTERS INDIAN COAST GUARD • NATIONAL TECHNICAL
RESEARCH ORGANISATION (NTRO)

TECHNICAL EVIDENCE DOSSIER & STATUTORY NOTICE OF VIOLATION
ISSUED UNDER SECTION 356E, MERCHANT SHIPPING ACT 1958 & MARPOL 73/78 ANNEX I

Dossier Ref:	ICG-MEP-DOSSIER-20260905-GULF	Date of Issue:	05 Sep 2026, 15:56 UTC
Maritime Sector:	Gujarat Offshore / Northern Arabian Sea	Legal Classification:	ILLEGAL DISCHARGE (CRIMINAL)
Target Vessel:	MT NEPTUNE GLORY	IMO / MMSI:	9384721 / 636019482
Flag State:	Liberia	Attribution Certainty:	86.7% (CONFIRMED)

1. SATELLITE RADAR REMOTE SENSING (SAR) OBSERVATION

Sensor Platform:	Sentinel-1A C-Band SAR	Acquisition Timestamp:	2026-09-04 13:10:00 UTC
Swath Mode & Pol:	Interferometric Wide (IW) Swath / VV (Co-polarization optimal for sea surface roughness)	Ground Resolution:	10.0 meters/pixel
Observed Slick Centroid:	22.4650° N, 69.2151° E	Total Spill Surface Area:	22.86 km²
Estimated Discharge Mass:	502.8 Metric Tonnes	Lookalike Rejection Score:	65.0% Mineral Crude Confidence
ADIOS Weathering State:	Heavy Chocolate Mousse (Highly Viscous Emulsion)	Evaporative Loss / Mousse:	25.7% Evaporated / 75.0% Water Uptake

2. REVERSE LAGRANGIAN HYDRODYNAMIC DRIFT ANALYSIS (ORIGIN ATTRIBUTION)

Using an advection-diffusion Lagrangian numerical formulation integrated with real-time oceanographic current vector fields ($\Delta u=0.28$ m/s, $\Delta v=0.18$ m/s) and Ekman surface windage (3.2% Stokes drift factor with 15° Coriolis deflection), the slick plume trajectory was hindcasted backwards in time. The particle convergence analysis resolved the **exact discharge origin point**:

Spill Origin Point (x_0, y_0):	22.35227° N, 69.09193° E	Calculated Slick Age:	10.5 Hours prior to SAR pass
Estimated Discharge Time:	T – 10.5h (05 Sep 2026 approx 02:40 IST)	Total Hydrodynamic Drift:	17.84 km displacement

3. AIS CORRELATION & KINEMATIC BEHAVIORAL ANOMALIES

Analysis of **6 vessels** operating in the sector filtered down to **5 corridor candidates**. Vessel **MT NEPTUNE GLORY** exhibited definitive spatio-temporal intersection with the discharge epicenter:

Evaluation Metric	Observed Telemetry Value	Anomaly Assessment	Score Weight
Closest Point of Approach (CPA)	0.86 NM (1.59 km)	Exact spatial co-location with slick root	89.2/100
Temporal Coincidence	$\Delta t = 0.50$ hours from origin	Precise temporal synchronization at t_0	91.7/100
Speed Over Ground (SOG)	Drop: 14.7 → 5.1 kts	Severe speed drop (14.7 -> 5.1 kts) matching illegal bilge purge profile	100.0/100
Vessel Risk Category	Crude Oil Tanker (DWT: 158,000 T)	High capacity oily water / slop separator	95.0/100

5. COASTAL HAZARD & BEACHING INTERCEPTION ALERT

CRITICAL SHORELINE ALERT: Hydrodynamic forecasting indicates slick impact along the coastline within 8.0 hours. Interception coordinates: 22.5649° N, 69.3182° E. Immediate mobilization of ICG Pollution Response (PR) Vessels and containment booms is ordered.

5. STATUTORY DIRECTIVES & DETENTION ORDER

NOTICE TO MASTER, OWNER, AND FLAG ADMINISTRATION:

Pursuant to powers vested under Section 356J and Section 356E of the Merchant Shipping Act, 1958:

- 1. **PORT STATE CONTROL INTERCEPTION:** The Principal Officer, Mercantile Marine Department (MMD), is directed to **DETAIN** vessel **MT NEPTUNE GLORY (IMO: 9384721)** upon entry into any Indian port or anchorage until mandatory oil record book (ORB Part I & II) inspection and bilge manifold swab sampling are concluded.
- 2. **FINANCIAL SECURITY:** A maritime environmental lien of **INR 25,00,00,000 (Twenty-Five Crore Rupees)** is hereby levied to cover offshore containment, dispersant spraying, and coastal remediation costs.
- 3. **FLAG STATE NOTIFICATION:** A copy of this satellite radar and AIS kinematic forensic package is dispatched to the International Maritime Organization (IMO) and the Maritime Administration of Liberia.

Forensic Chain-of-Custody SHA-256 Hash:
acbde85bf7f931d6ea33737d8b37bbe1c506d90794e8b491ddca53ff04d6d72a

Authorized Signatory:
Inspector General (Operations & MEP)
Indian Coast Guard Headquarters, New Delhi